

ELECTRIC CHARGES

Electric Charge

Definition:

Electric charge is the basic property of matter due to which it experiences electrical force.

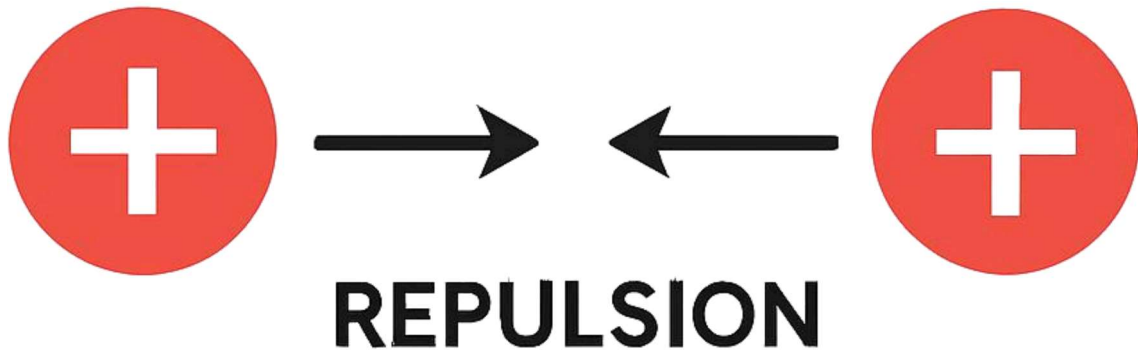
Types of Charges:

- Positive charge
- Negative charge

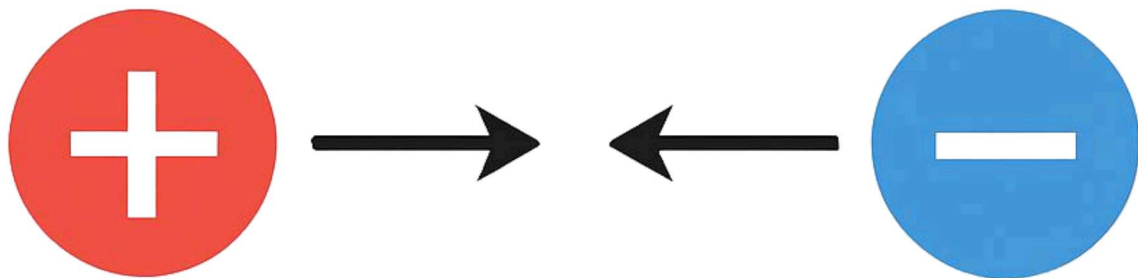
Properties of Electric Charges

- Like charges repel each other.
- Unlike charges attract each other.
- Charge is conserved.
- Charge is quantized.

LIKE CHARGES



ATTRACTION



UNLIKE CHARGES

Quantization of Charge

$$q = ne$$

Where:

- q = Charge

- n = Integer
- e = Electronic charge

Conservation of Charge

Electric charge can neither be created nor destroyed.

Coulomb's Law

Statement:

The electrostatic force between two point charges is directly proportional to the product of charges and inversely proportional to square of distance between them.

Formula:

$$F = kq_1q_2 / r^2$$

Where:

- F = Electrostatic force
- q_1, q_2 = Charges
- r = Distance between charges
- k = Electrostatic constant

SI Unit of Charge

Coulomb (C)

Electric Field

Definition:

The region around a charge where another charge experiences force is called the electric field.

Electric Field Intensity:

$$E = F / q$$

SI Unit:

N/C

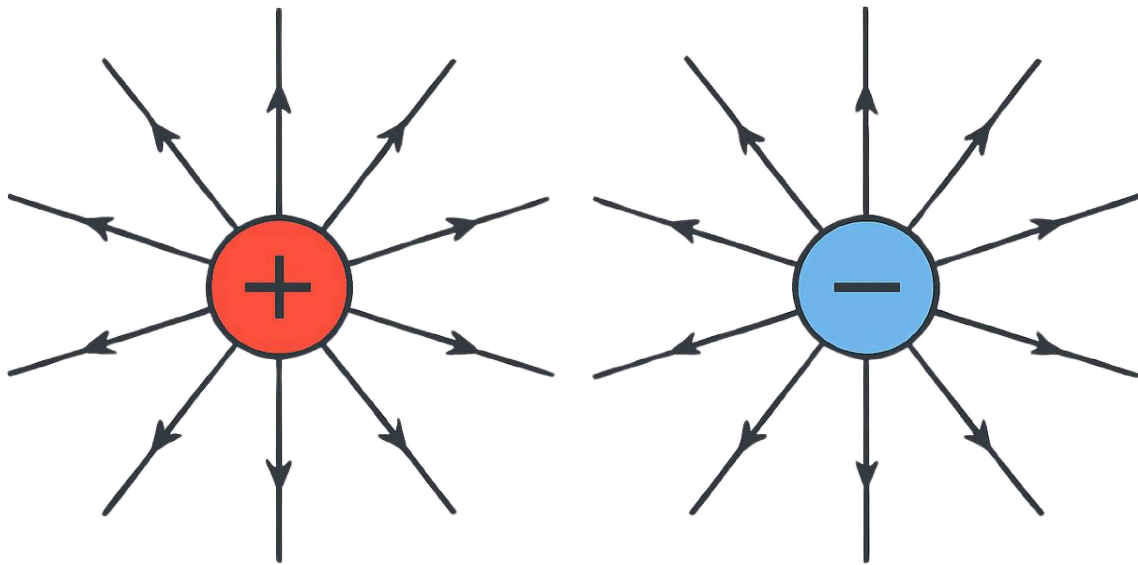
Electric Field Lines

Definition:

Imaginary lines representing the direction of an electric field are called electric field lines.

Properties:

- Field lines start from positive charge.
- Field lines end at negative charge.
- Field lines never intersect.
- Closer lines indicate stronger fields.



ELECTRIC FIELD

Electric Dipole

Definition:

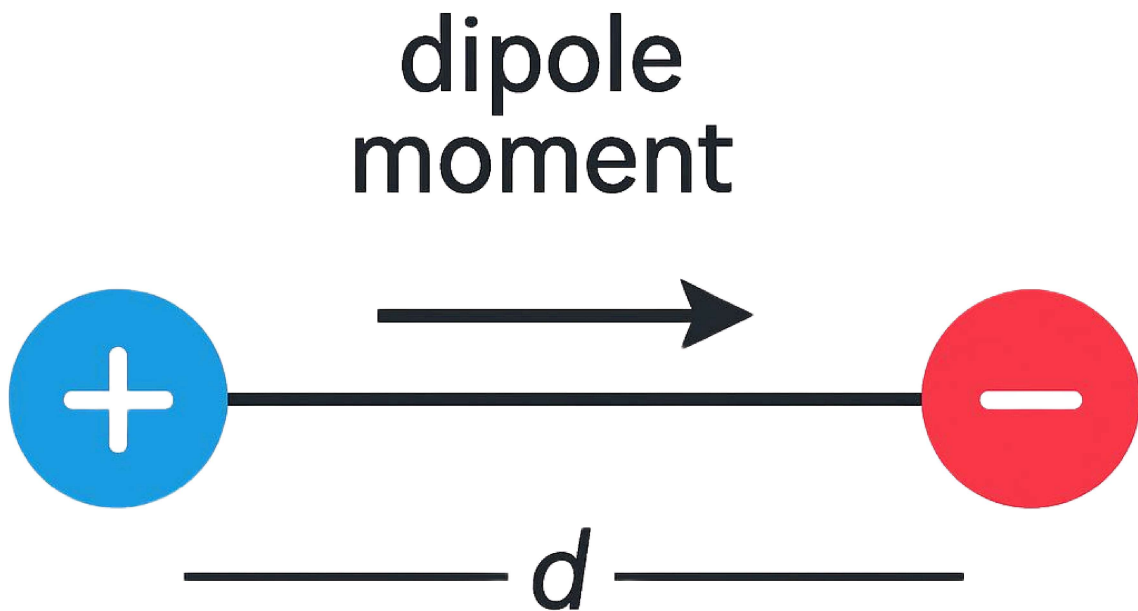
A pair of equal and opposite charges separated by a small distance is called an electric dipole.

Dipole Moment:

$$p = q \times 2l$$

SI Unit:

C·m



ELECTRIC DIPOLE

Applications of Electric Field

- Electrostatic precipitators
- Inkjet printers
- Photocopiers

Quick Revision Points

- Electric charge is the basic property of matter.
- Like charges repel and unlike charges attract.
- Charge is conserved and quantized.
- Coulomb's law gives electrostatic force.
- An electric field exists around charges.
- Electric field intensity is $E = F / q$.
- An electric dipole consists of equal and opposite charges.
- Dipole moment is represented by p .

Important Formulae

- $q = ne$
- $F = kq_1q_2 / r^2$
- $E = F / q$
- $p = q \times 2l$

Conclusion

Electric charges and electric fields form the foundation of electrostatics.